Muhammad Muizz Zafar

muizzafar11@gmail.com · 905-805-9142 · linkedin.com/in/muizz-zafar · github.com/muizzzafar11 ·

Software Developer | Backend Systems | Testing, Debugging & Reliability

Technical Skills

Programming Languages:	C++, C#, Python, Java, JavaScript, TypeScript, GoLang, SQL, HTML/CSS
Frameworks/Tools:	NUnit, Google Test, Cypress, Jenkins, AWS, Docker, Vue.js, React, Angular, Git

Work Experience

Software Developer Intern — Sept 2024 - Present
Kinaxis Inc. — Ottawa, ON

- Developed and maintained production code for the Maestro supply chain algorithms planning platform using **C++** and **C#**.
- Designed automated **unit and integration tests** using Google Test and NUnit to validate complex planning algorithms.
- Implemented **C++ services** supporting scope control to detect incompatiblities between different features of the Rapid Response product which was used by customers and reduced L3 escalations by customers.
- Designed automated queries in SQL to detect **50+** known behavioral differences across different algorithms between the legacy and modular codebase.
- Integrated automated test workflows into **Jenkins CI/CD pipelines** to support reliable continuous delivery.
- Used **Javascript** to write automated workflows within the rapid response product to run the known difference queries and visualize the known differences for directors, VP, and customers to see the differences between legacy and modular codebase
- Participated in code reviews and documented testing workflows to maintain high code quality and improve team knowledge sharing.

Full Stack Developer — Sept 2023 - Aug 2024
University of Ottawa — Ottawa, ON

- Developed backend and frontend features for the University emergency alert system using **Laravel**, **Drupal**, and JavaScript.
- Built and maintained REST APIs and optimized database queries to improve response times for a platform serving **20,000+ users**.
- Implemented automated **unit and end-to-end tests** using Cypress, Jasmine, and JUnit to ensure application reliability.
- Worked closely with cross-functional teams to debug production issues and deliver stable releases.
- Investigated and resolved production issues by debugging backend APIs and database queries, improving system reliability.

Cloud Developer Intern — May 2023 - Aug 2023 & Sept 2022 - Dec 2022
Trend Micro Inc. — Ottawa, ON

- Developed backend services and serverless infrastructure using **Go**, **AWS Lambda**, and **Athena** to support scalable cloud systems.
- Built monitoring workflows with **CloudWatch** and **Splunk** to detect system errors and improve system reliability.
- Implemented automated integration tests to validate serverless pipelines handling high-volume cloud artifacts.

Full Stack Developer Intern — Jan 2022 - Apr 2022
Applied Mind Inc. — Ottawa, ON

- Developed backend services using **Python**, **GraphQL**, and **PostgreSQL**.
- Built monitoring dashboards in **Vue.js** for real-time validation of system events.
- Automated validation workflows to reduce manual QA verification and improve system observability.

Full Stack Developer — Jun 2019 - Jun 2021
Tech Under Twenty — Oakville, ON

- Developed RESTful APIs using **.NET Core (C#)** supporting user and member management services.
- Built frontend components with **Angular** and debugged UI/API integration issues.
- Collaborated with developers to review code, troubleshoot issues, and deliver stable application features.

Education

University of Ottawa — Ottawa, ON
Bachelor of Applied Science, Software Engineering *GPA: 3.7/4.0* — September 2020 - March 2025